Enhancing Stock Prediction with Sentimental and Relational InformationDistilled from Large Models

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Abstract

An increasing number of investors are dedicating their efforts to enhancing the accuracy of stock market predictions, a crucial factor in financial markets that aids in making well-informed decisions and optimizing their investment strategies. The development of deep learning methods has evolved from solely considering fundamental price information to a broader spectrum of information, such as sentiment features from news articles. However, these methods often struggle to effectively extract such features when multiple companies are mentioned in a single news article (more details can be referred to Section 1). With the advancement of large language models (LLMs), which can more precisely extract the knowledge people require from text. To overcome the shortcomings of previous work, we propose a novel distillation method for precise knowledge extraction, including sentimental and fine-grained relational knowledge. After that, we incorporate the knowledge as augmented features to assist in the predictions of stock forecasting models. Through experiments on the real Chinese stock market, CSI300 and CSI500, our proposed method can indeed lead to significant improvements in the performance of stock prediction models, demonstrating a performance improvement of baseline models up to 3.23% in terms of accuracy.

CCS Concepts

• **Information systems**; • **Information systems applications**; • **Data mining**; • **Data stream mining**;

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Keywords

Large language models, Stock prediction, Knowledge distill, Deep learning

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1 Introduction

Quantitative investment research is increasingly utilizing deep learning for stock prediction, with its ability to automate intricate feature extraction from vast datasets [6, 19]. These modern methods mainly use historical price data, including opening, closing, highest and lowest prices, and trade volume, to enhance prediction accuracy [2, 22]. The growing reliance on deep learning highlights its potential to transform traditional investment tactics. As stock prediction remains central, the overlap of deep learning and quantitative investment continues to be a major research interest.

While current methodologies have made progress, they differ from the holistic approach of human traders who consider factors beyond historical prices, such as significant news events [7, 18]. Recent research trends emphasize multi-modal data, exploring sentiments and relationships in news articles related to stocks. Traditional sentiment analysis often relies on dictionaries like the LM dictionary [5] and VADER [16, 20]. However, these tools have limited vocabulary and lack context understanding, possibly affecting accuracy [17]. There's a growing emphasis on creating sentiment extraction methods that better resonate with human trading decisions.

Large Language Models (LLMs) have excelled in various natural language processing tasks, largely due to their transformer block and attention mechanisms which prioritize key input segments [10]. In finance, LLMs analyze news data, focusing on sentiments [17] and relationships [4] crucial for accurate stock prediction. However, the "hallucination" issue in LLMs [1], depicted in Figure 1, can compromise accuracy. Hence, reliable sentiment extraction

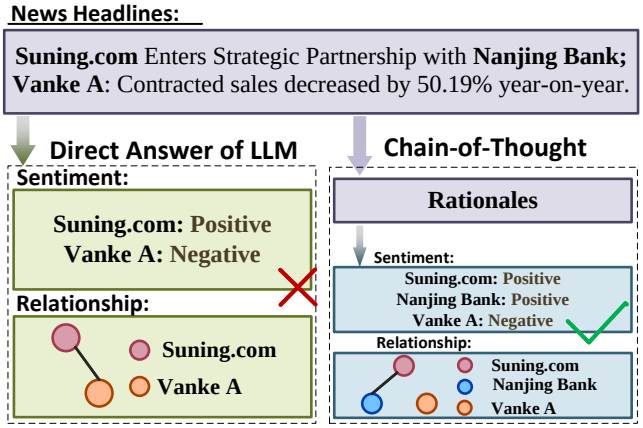

Figure 1: Direct prompts with LLMs can lead to the hallucination problem and less accurate results. However, using a chain-of-thought approach offers a reasoned process for more accurate outcomes, as illustrated in Figure 2.

and relationship identification are vital for maximizing LLMs' stock prediction potential.

Building on chain-of-thought (CoT) prompts in LLMs [10, 14, 26, 29], we present Chain-of-Thought-Aided Distillation (CoTAD). CoTAD employs chain-of-thought to distill sentiment and relational knowledge, capturing market sentiments and company interplays, thereby refining predictions in smaller models. This knowledge is converted to multi-hot embeddings and relation graphs, then fused with historical prices, enhancing stock prediction. In summary, CoTAD combines chain-of-thought reasoning and feature augmentation to elevate stock prediction accuracy. Our contributions include:

- We employ the chain-of-thought method to enhance the precise extraction of news-related insights, capturing both the sentiment and relational aspects of stock information.
- A framework utilizing knowledge distillation and LLMs is introduced to enhance the accuracy of compact stock prediction models.
- Through experiments on two real-world Chinese stock market data, we show that our proposed approach significantly boosts the performance of various stock prediction models, with improvements up to 3.23%.

2 Related Work and Preliminary

Stock movement prediction can be framed as a binary classification problem, with "upward" and "downward" labels based on closing prices ($y_t^i = 1$ or $y_t^i = 0$). Models can be non-graph-based or graph-based. The former views stocks independently and may use RNNs to capture temporal price dependencies. Attention-based RNNs like [22] extract signals and states, while [8] introduces adversarial training. The formula for these methods is:

$$\hat{y}_t^i = f([\mathbf{x}_{t-1}^i, \mathbf{x}_{t-2}^i, ..., \mathbf{x}_{t-T+1}^i], S; \Theta), \tag{1}$$

with $\mathbf{x}_t^i \in \mathbb{R}^D$ denoting D historical features, T as the time lag, Θ as parameters, and S as other data like sentiment. When no other features are considered, S can be left out. We use the cross entropy as the loss:

$$\mathcal{L} = -\left[y_t^i \log(\hat{y}_t^i) + (1 - y_t^i)\log(1 - \hat{y}_t^i)\right]. \tag{2}$$

The stock market can be seen as a complex network where stocks interrelate. Graph-based methods use this structure, leveraging GNNs to aggregate stock features and improve predictions. Some utilize market knowledge like industry data for predefined graphs. [3] models shareholding relations in a corporate graph and employs GCNs for node classification. [9] uses industry and Wikipedia relations to craft a graph, enhancing GCNs with LSTMs. [21, 28] creates multiple graphs to capture industry, shareholding, and news relations, revealing cross effects. [5] fuses price and sentiment data, constructing relations for feature aggregation. The formula for these graph-based approaches is:

$$\hat{\mathbf{Y}}_t = f([X_{t-1}, X_{t-2}, ..., X_{t-T+1}], S, R; \Theta), \tag{3}$$

where $X_t \in \mathbb{R}^{N \times D}$ indicates historical price features of N stocks, R is the relational information among stocks and $\hat{\mathbf{Y}}_t$ is the predictive results of N stocks. Thus, the loss function is as follows:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_t^i \log(\hat{y}_t^i) + (1 - y_t^i)\log(1 - \hat{y}_t^i)\right]. \tag{4}$$

3 Methodology

3.1 Framework

The proposed framework leverages the analytical strengths of Large Language Models (LLMs) to boost prediction accuracy for stock movements, allowing for training of refined models on a reduced scale. As illustrated in Figure 2, our method unfolds in two phases: Firstly, using LLMs with news articles, we extract deeper stock insights beyond just historical price data. This stage yields augmented training samples that fine-tune smaller stock prediction models. Next, this enriched stock knowledge aids in training these compact models, amplifying their forecasting accuracy. In the prediction phase, these models utilize the acquired insights to predict stock price movements.

3.2 Knowledge Distillation

Previous studies have highlighted the benefits of combining sentiment analysis and news-based relational data for stock predictions [4, 5, 11, 17, 28]. Given the LLMs' ability to provide explanatory reasoning [14, 26], we employ the Chain-of-Thought (CoT) method to craft LLM prompts for sentiment and relational analysis. This aims to guide the LLM through a structured deduction process, allowing for deeper understanding. The structured prompt we developed follows:

Rationale Prompt: Forget all your previous instructions. I want you to act as an experienced financial expert. I will offer you a news headline. Your task is to analyze the influence of the news on these stocks: [firms list]. The output example of JSON format: Rationale list: firm 1: rationale, firm 2: rationale. The news headline is as follows: [news headline].

The [list of companies] in the prompt refers to businesses mentioned in the news. As seen in Figure 2, the rationale covers a wide

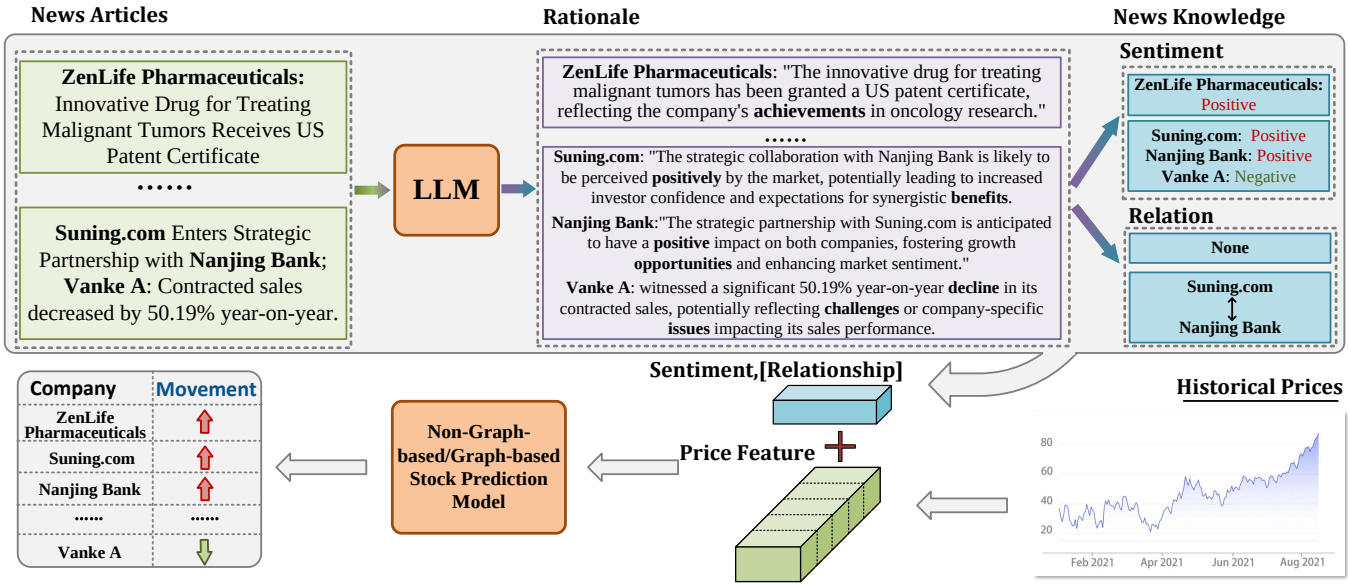

Figure 2: The framework of CoTAD. We first utilize chain-of-thought prompting to extract sentiment and relationship knowledge. After that, we utilize the extracted knowledge as augmented features to guide the training of the smaller stock prediction model.

range of analysis, including achievements and positive outcomes. This enhances the extraction of sentiments and relationships from the news. We then extract sentiment and relational knowledge using news headlines and the provided rationales. Using the chain-of-thought approach [26], we analyze rationales in stock news, aiding the extraction of key knowledge for streamlined stock prediction models. In this paper, the LLM-derived knowledge is divided into two main areas: sentiment analysis and relational data.

Sentimental Knowledge Prompt: I will offer you a news headline and the corresponding rationales, your task is to: 1. Identify the sentiments of the firms: positive, negative, or neutral. 2. Only consider these firms:[firm list]. 3. Sentiment Examples:{few_shot_learning_examples}. The news headline and the corresponding rationale are as follows: [news headline, rationale].

Relational Knowledge Prompt: I will offer you a news headline and the corresponding rationales, your task is to: 1. Determine which firms are correlated and according to the news headline and the rationales. 2. Only consider these firms: [firms list]. 3. Relationship Examples:{few_shot_learning_examples}. The news headline and the corresponding rationales are as follows: [news headline, rationale].

For each time step, we assign a three-dimensional sentiment vector $\mathbf{st}^i \in \mathbb{R}^3$ to each firm, representing positive, negative, and neutral sentiments. For example, a positively viewed firm would be represented as (1,0,0). Concurrently, we construct the graph $G^t_{news} = (V, E^t)$ with firms as nodes. Edges are added between firms when a strong relationship is detected by the LLM.

3.3 Feature Fusion

We use extracted sentimental and relational knowledge as augmented features to improve model predictions. For sentiment knowledge, a feature fusion layer integrates historical feature, $\mathbf{x}^t_i$, with sentiment feature, $\mathbf{st}^t_i$, forming a new input feature. Using the method from [5], we analyze interactions between $\mathbf{x}^t_i$ and $\mathbf{st}^t_i$ through vector-matrix-vector (VMV) products. The features are then concatenated and transformed linearly. The fusion is represented as:

$$\mathbf{f}^t_i = \tanh(\mathbf{x}^t_i \mathcal{F}_i^{[1:F']} \mathbf{st}^t_i + W_f[\mathbf{x}^t_i \parallel \mathbf{st}^t_i] + \mathbf{b}_f), \tag{5}$$

where $\parallel$ is concatenation and both W_f and $\mathbf{b}_f$ are learnable parameters. For the relational graph, an attention-based graph fusion method is employed before the first GNN layer, merging the news relational graph with the original one:

$$G = \alpha \times G_{origin} + \beta \times G_{news}. \tag{6}$$

4 Experimental Evaluation

4.1 Data and Metrics

We utilize two real market datasets from the Chinese CSI300 and CSI500 index to assess our methodology, encompassing price and news data between March 1, 2019, and October 1, 2023. Price data is obtained from Tushare, while 143,396 relevant news articles were extracted from JRJ.com after filtering from a total of 1,356,756 articles. The CSI300 and CSI500 dataset includes 1154 trading days. Price data, normalized using Z-score, contains attributes like opening, closing, high, and low prices, as well as trading volume. Additionally, the news data is processed to extract sentimental and relational features. The datasets are split as follows: 80% for training, 10% for validation, and 10% for testing. For evaluation, we employ accuracy

Table 1: Performance compared with seven stock prediction baselines.

Methods		baseline performance		Sentiment Aided		Relation Aided		Sentiment & relation Aided	
		ACC(%)	MCC	ACC(%)	MCC	ACC(%)	MCC	ACC(%)	MCC
CSI300	GCN [13]	50.49	0.0087	51.32	0.0170	51.50	0.0127	53.72	0.0322
	GAT [24]	50.78	0.0051	51.57	0.0137	52.77	0.0105	53.28	0.0441
	TGC [9]	51.95	0.0283	52.73	0.0355	52.49	0.0292	53.70	0.0342
	ADGAT [5]	51.45	0.0381	52.16	0.0468	52.53	0.0412	53.46	0.0573
	MGAR [23]	52.07	0.0424	52.65	0.0427	53.38	0.0523	54.19	0.0613
	VGNN [27]	53.06	0.0511	54.01	0.0564	52.95	0.0447	55.72	0.0404
	MDGNN [21]	52.19	0.0316	53.86	0.0382	53.67	0.0561	55.34	0.0605
CSI500	GCN [13]	51.06	0.0123	51.78	0.0195	51.68	0.0141	53.03	0.0404
	GAT [24]	50.88	0.0134	52.66	0.0176	52.23	0.0202	54.75	0.0673
	TGC [9]	51.92	0.0265	53.80	0.0295	52.64	0.0276	53.05	0.0463
	ADGAT [5]	52.68	0.0317	52.84	0.0365	53.37	0.0371	54.15	0.0582
	MGAR [23]	53.57	0.0487	54.69	0.0527	54.01	0.0417	55.66	0.0664
	VGNN [27]	53.04	0.0222	54.24	0.0459	55.06	0.0326	55.85	0.0502
	MDGNN [21]	52.47	0.0323	53.01	0.0469	52.56	0.0374	55.39	0.0488

(ACC) and Matthews Correlation Coefficient (MCC), in line with prior stock prediction studies [15, 25].

To evaluate the performance of our method, we conduct experiments comparing it with multiple GNN-based stock prediction methods, including GCN, GAT, TGC, ADGAT, MGAR, VGNN, and MDGNN. The proposed method is integrated with each predictor to enhance predictive performance. All model parameters are trained using the Adam optimizer [12] on a single NVIDIA RTX 4070Ti 12G GPU. The learning rate is set to 5e-4, and the batch size is set to 64. Each experiment is independently repeated five times, with the mean reported.

4.2 Experimental Results

Our experimental results, detailed in Table 1, highlight significant enhancements when using the chain-of-thought-based distillation methods compared to baseline models. Integrating sentimental and relational knowledge individually resulted in accuracy improvements of up to 1.88% and 2.02%, respectively. The combined approach achieved the greatest improvement, with an accuracy increase of up to 3.23%.

The chain-of-thought methodology crucially enhances the LLMs' capability in generating news knowledge, ensuring contextual coherence. This method greatly aids in translating news text into meaningful analytical insights. Sentimental knowledge, in particular, grants a deeper understanding of market dynamics, with prior research validating news sentiment's predictive power over stock price movements. Additionally, the inclusion of fine-grained relational attributes allows models to understand intricate interdependencies among companies, leading to more precise predictions. The combined use of sentimental and relational knowledge yields the highest performance. This indicates that both types of knowledge complement each other, leading to a more holistic understanding of the news data. This synergistic effect underscores the importance of integrating multiple types of knowledge for optimal performance.

Our findings underscore the significance of the chain-of-thought approach in LLMs for synthesizing news knowledge. Both sentimental insights and detailed relational features play crucial roles in enhancing the accuracy of stock prediction models. They provide a nuanced understanding of market sentiments and help decipher the multifaceted factors influencing stock prices. This methodology not only improves the models' performance but also ensures that the generated insights are contextually rich and analytically robust, leading to more informed and accurate stock predictions.

Table 2: Ablation experiments for verifying the effectiveness of using CoT, zero-shot and few-shot.

Methods	non-CoT	zero-shot-CoT	few-shot-CoT
GCN	51.57%	53.41%	53.72%
GAT	50.98%	52.46%	53.28%
TGC	52.39%	53.22%	53.70%
ADGAT	53.24%	53.47%	53.46%
MGAR	53.85%	54.05%	54.19%
VGNN	53.67%	54.08%	55.72%
MDGNN	53.16%	53.67%	55.34%

4.3 Ablation Experiments

Table 2 showcases the impact of the Chain-of-Thought (CoT) methodology in both zero-shot and few-shot contexts on CSI300 dataset. MDGNN's performance is noteworthy, improving from 53.16% without CoT to 53.67% with zero-shot CoT, and reaching 55.34% in the few-shot CoT setup. Omitting any CoT component significantly diminishes MDGNN's performance. Comparable improvements are observed in other methods like GCN, GAT, and TGC upon CoT integration, especially in zero-shot and few-shot contexts. In summary, CoT substantially boosts the efficacy of various models, exemplifying its pivotal role in enhancing predictive accuracy.

5 Conclusion

In this paper, we propose a novel knowledge distillation technique based on the chain-of-thought paradigm to enhance stock prediction models. Unlike traditional methods requiring resource-intensive fine-tuning, our approach leverages the inferential capabilities of Large Language Models (LLMs) for efficient knowledge extraction from news articles. This strategy surpasses conventional sentiment extraction tools like LM dictionaries. By integrating knowledge from ChatGPT, we improve the performance of stock

prediction models without the computational overhead of extensive fine-tuning. Our method offers an innovative pathway for enhancing predictive accuracy in the financial domain, highlighting its potential for wider application in quantitative research.

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